

Latency in Distributed LLM Inference

JULY 14, 2026

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Thesis Scope:
Communication Latency in Distributed LLM Inference

Research Focus: How does communication latency affect distributed LLM inference performance across GPU interconnect topologies and tensor-parallel configurations?

- Single-GPU phase focuses on pure model / hardware generation behavior
- Multi-GPU phase examines distributed models across various GPU transports / interconnects and injects controlled latency

Experiment Set-Up:

- vLLM 0.11.2
- LLaMa models (3.2-1B, 3.1-8B, 2-13B) – dense transformers
- V100-32GB (connected via PCIe) and A100-40GB GPUs (via NVLink)
- Main metrics: time-to-first-token (TTFT), inter-token latency (ITL), output throughput, energy cost per token

Single-GPU Baselines – KV Cache Saturation

Set-up:

- A100 vs V100 using Llama-3.1-8B
- Swept input/output length and concurrency: to isolate prefill/decode behavior and isolate batching/queueing effects
- Estimate KV cache capacity using vLLM's internal allocation logic:

$$\text{KV cache available} = (\text{VRAM} \times \text{gpu_mem_util}) - \text{model weights} - \text{activations/overhead}$$

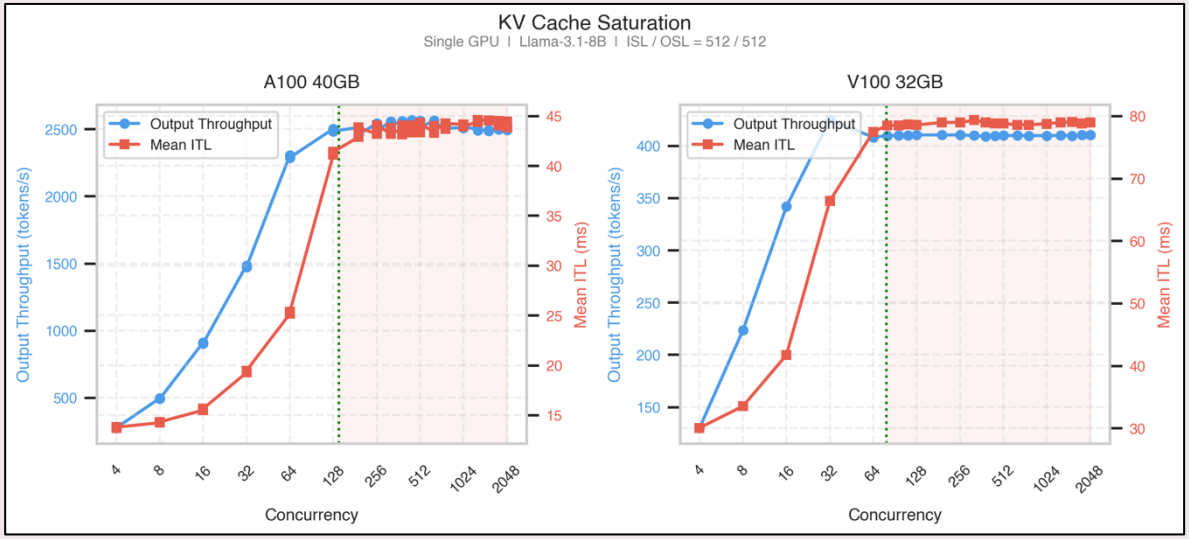
$$\text{KV cache bytes/token} = 2 \times \text{num layers} \times \text{num kv heads} \times \text{head dim} \times \text{dtype bytes}$$

  - Activations/overhead assumed to be 5-10% of total VRAM.

Findings:

- TTFT scales linearly with input length, output length has negligible effect
- ITL and throughput increase with concurrency, then plateau near predicted KV cache saturation points

KV Cache Saturation on Decode Performance



- Both decode phase metrics plateau after cache saturation, GPU reaches a steady state processing incoming requests

Scaling to Multi-GPU: Transport & Latency

Setup:

- TP = 1,2,4 for two server topologies (A100/NVLink and V100/PCIe)
- Transport comparison:
  - default (best available – NVLink if present)
  - shared memory (peer-to-peer disabled – over PCIe)
  - network socket
- Controlled latency injection into all-reduce communication calls
  - investigate how performance degrades as communication latency increases, looking at latency metrics (TTFT, ITL), throughput, and power usage

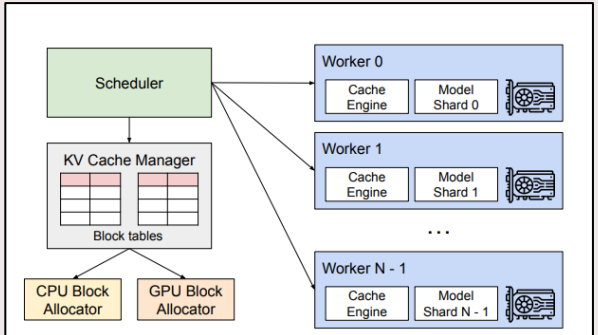
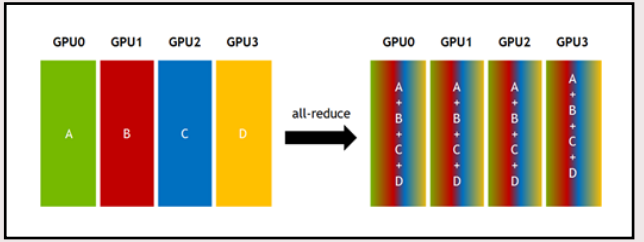


Figure 4. vLLM system overview.

Source: vLLM paper



Source: NVIDIA technical blog

Multi-GPU Scaling & Transport Bottlenecks

Scaling to more GPUs:

- optimal TP distribution depends on model size, hardware, type of data transport
- Memory and compute benefits of scaling across more GPUs don’t always outweigh extra communication cost

Transport & Peer-to-Peer (P2P)

- P2P memory access over NVLink allows for efficient scaling
- When forced to fall back to SHM over PCIe or network sockets, communication overhead quickly degrades performance

Injected Latency & Power Profiles

Adding latency within all-reduce - the communication call GPUs use to aggregate partial activations during forward passes

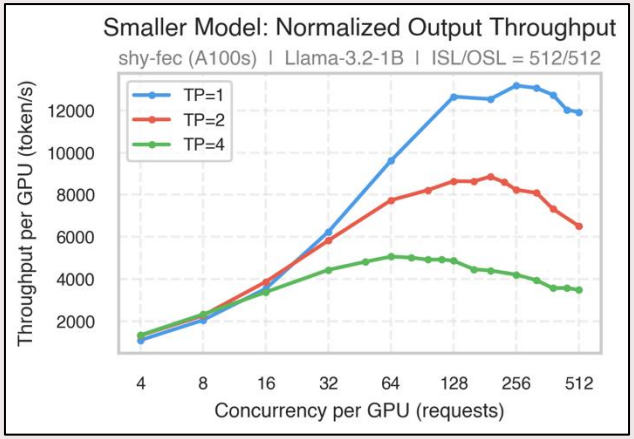
How does a distributed LLM respond to additional link latency?

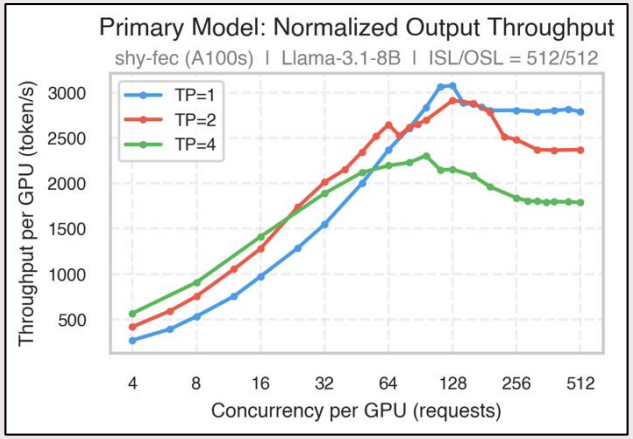
- TTFT, ITL, energy cost per token all scale linearly with increase in all-reduce latency, throughput decreases sub-linearly
- More GPUs means more sensitivity to communication latency
- KV cache saturation impacts prefill performance, forcing requests to queue, leading to much higher TTFT

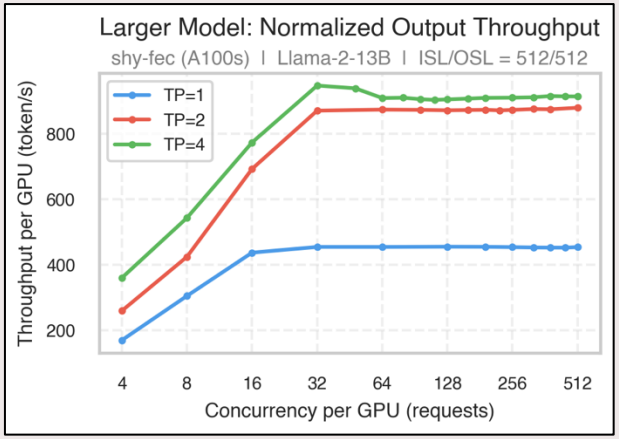
Currently: analyzing power profiles, estimating scaling performance by extrapolation, possibly running larger experiments on Spike-1

Thank you!

Scaling with Models of Different Sizes







- Small models don't benefit from TP due to communication overhead
- With NVLink, TP=4 in optimal for 13B -> memory and compute benefits outweigh communication cost
- How well an LLM scales relies directly on the size of the model, available VRAM, and type of data transport

TU/e